

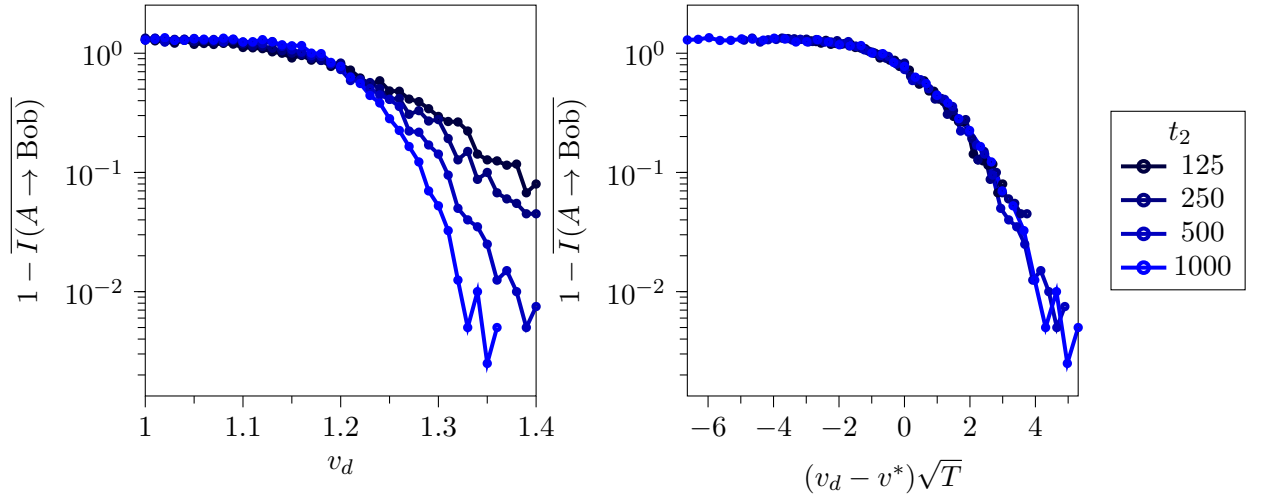


Figure 9: Left: Information phase transition around critical point. Right: Data collapse under a finite-time rescaling. Here we set $t_1 = 100$ and $p = 0.1$. This is the result of an average over 400 disorder realizations.

5.2 Teleportation protocol

Here we make use of a fundamental property of bipartite entropies to derive a distinct protocol which exhibits an equivalent transition in coherent information but for a channel complementary to the previous protocol. In this new protocol Alice carries out an equivalent role i.e. she prepares the impurity qubit, I , in a specific state, allows it to interact with the medium, M , and then replaces it with a new qubit, R , in an infinite temperature state. Throughout this the total evolution is unitary, as such any information which Bob cannot access from the discarded qubit, D , or the final state of M must be accessible elsewhere in the system. With this in mind we now introduce a third party, Charlie, who is also attempting to deduce information about the initial state of the impurity. Charlie has qubits that are maximally entangled with the initial states of M and R , A_M and A_R respectively, as well as having access to the final state of R . These new ancilla, A_M and A_R , play a similar role to A in that they perfectly remember the initial state of the medium and the new qubit. Charlie having access to these implies he can make use of this information to help decode Alice's information. However, unlike Bob, Charlie has no knowledge of the discarded qubit, D , or the final state of M . The degrees of freedom Charlie can access are complementary to those accessible by Bob. This allows us to greatly simplify the calculation of the coherent information available to Charlie.

For pure states, like the one describing the quantum correlations between Alice, Charlie and Bob, all bipartite entanglement entropies satisfy

$$S(\rho_B) = S(\rho_{B^\perp}) \quad (12)$$

where $B^\perp$ is the complement to the subspace B . Take a generic tripartite space consisting of $X \cup Y \cup Z$ and consider the coherent informations $I(X \rightarrow Y)$ and $I(X \rightarrow Z)$. Making use of Eq. (12) allows us to relate these in the following way:

$$\begin{aligned} I(X \rightarrow Y) &= S(\rho_Y) - S(\rho_{X \cup Y}) \\ &= S(\rho_{Y^\perp}) - S(\rho_{(X \cup Y)^\perp}) \\ &= S(\rho_{X \cup Z}) - S(\rho_Z) \\ &= -I(X \rightarrow Z). \end{aligned} \quad (13)$$